

Pavel Golikov

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PROFESSIONAL PROFILE

AI Researcher and agent engineer specializing in robust machine reasoning, retrieval-augmented generation (RAG), LLM evaluation, and alignment. Builds grounded agentic workflows, vector-retrieval systems, and dataset generators for context-management evaluation in LLM agents. Former Military Intelligence Operator who held a Top Secret clearance, bringing a threat-modeling mindset to AI security and model capability evaluation.

TECHNICAL SKILLS

Languages: Python, C++, Java, SQL, LaTeX

AI & Agents: PyTorch, LangGraph, LangChain, RAG, Hugging Face, vLLM, Transformers, Gemini/OpenAI/Anthropic APIs

RAG & Evals: Vector storage, embeddings, BM25, hybrid retrieval, cross-encoder reranking, MRR/nDCG/recall, groundedness, validation

Systems: Linux, SQLite, GitHub Actions CI, AWS, Apache Flink, Distributed GPU Clusters

AI RESEARCH & AGENT ENGINEERING

- **ArbiGraph** | [arXiv](#) | [GitHub](#) | [Hugging Face](#) 2026 – Present
 - Built an open-source benchmark and dataset generator for evaluating context management in tool-assisted LLM agents across arbitrarily scalable, long-horizon task graphs.
 - Generated math, word-problem, and Python-tracing workflows with executable ground truth, enabling exact evaluation of state retention, updates, propagation, and stale-state reuse without manual labels.
 - Controlled graph length, branching, distractors, and value types; implemented a calculator-agent harness with answer extraction, tool-call validation, continuation handling, and repair prompts.
- **arXiv Research Agent** | [GitHub](#) | [Evals](#) 2026
 - Built an evaluation-first **RAG** literature-review agent with **LangGraph**, **LangChain**, **Gemini**, and **Chroma vector storage**; orchestrates arXiv search, PDF parsing, hybrid retrieval, reranking, and citation-grounded synthesis.
 - Benchmarked dense, BM25, hybrid, and cross-encoder retrieval on 50 hand-labeled questions over 570 chunks using MRR, nDCG, recall, and paired-bootstrap confidence intervals; evaluated groundedness and claim support.
 - Engineered deterministic map-reduce fan-out, retry-aware partial results, and resumable **SQLite checkpointing**; shipped 163 offline tests in CI and measured a median end-to-end cost of \$0.054 per run.
- **Robust Reasoning Benchmark (RRB)** | [arXiv](#) | [DOI](#) | [GitHub](#) 2026
 - Designed an adversarial evaluation framework leveraging 13 deterministic textual perturbations to decouple an LLM’s mechanical deciphering from its underlying mathematical logic.
 - Demonstrated that the attention drift occurs *within a single query’s Chain-of-Thought*, empirically showing that intermediate reasoning steps pollute the dense attention mechanism.
 - Engineered custom mechanistic interpretability pipelines in **PyTorch** for layerwise attention-allocation analysis across token index boundaries, testing models ranging from 7B to 30B parameters.
- **Fusing Adds and Shifts for Efficient Dot Products** | [IEEE CAL](#) | [GitHub](#) 2026
Hardware ML Research Toronto, ON
 - Proposed and validated a novel algorithmic optimization for dot-product computations, demonstrating a strong foundational understanding of hardware-level ML primitives and efficiency.

ENGINEERING & PROFESSIONAL EXPERIENCE

- **Graduate Researcher (PhD) — ML Systems & Agent Evaluation** 2022 – August 2026
University of Toronto, EcoSystem Research Group Toronto, ON
 - Conducted research spanning ML systems and efficient computation, with later work focused on robust machine reasoning, agent evaluation, context management, and AI alignment, in the EcoSystem Research Group and as a member of the Vector Institute.
 - Developed open-source evaluation frameworks, agentic RAG systems, and mechanistic-interpretability analyses, resulting in first-author research on context management and reasoning robustness.
- **Graduate Researcher (MSc Thesis) — Distributed Systems** 2020 – 2022
University of Toronto, EcoSystem Research Group Toronto, ON
 - Engineered a flexible IoT distributed data-streaming framework from scratch, designed to automatically partition computational streaming queries between edge devices and cloud instances.
 - Built the full software stack: programmed Arduino/C++ sensors for real-time biological data collection (EMG/ECG), developed custom socket networking protocols, and deployed cloud infrastructure using **AWS** and **Apache Flink**.
- **Intelligence Operator** 2013 – 2018
Canadian Armed Forces Canada
 - Formerly held a **Top Secret** security clearance while conducting rigorous analysis of classified information streams to produce actionable intelligence reports for command elements.
 - Developed a strong adversarial threat-modeling mindset, emphasizing operational security, rigorous data validation, and the identification of logical vulnerabilities in complex, multi-agent scenarios.
- **Mathematics Teacher** 2012 – 2013 & 2018 – 2019
Blyth Academy Toronto
 - Taught foundational mathematics to students in Grades 10, 11, and 12, developing the ability to distill and communicate complex quantitative concepts.

EDUCATION

- **University of Toronto** Toronto, ON
PhD in Computer Science (Leave of absence beginning September 2026) 2022 – Present
- **University of Toronto** Toronto, ON
Master of Science (MSc) in Computer Science 2020 – 2022
- **University of Toronto** Toronto, ON
Bachelor of Science (BSc) in Mathematics and Philosophy (Formal Logic) Graduated 2011